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Roll No

IT-7005(3)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Embedded System

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) Define embedded system and write the various categories of embedded system.
 - b) Explain challenges and issues of embedded systems in detail.
2.
 - a) Enumerate state machine model for the seat belt alarm system.
 - b) Discuss about the embedded system software performance analysis and optimization.
3.
 - a) Explain the concepts of sensors and actuators with suitable example.
 - b) Explain the three main processor technologies that are used with embedded system. Also gives the merits of each.
4. Explain the features of following register of 8051 micro controller
 - i) TCON
 - ii) SCON
 - iii) PCON
 - iv) PSW

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5.
 - a) Give instruction set completeness of 8085. Also write addressing modes.
 - b) Differentiate OS based and super loop based embedded firmware design approaches.
6.
 - a) Explain Task. Process and thread with their types and examples.
 - b) Discuss the various components and programming model used for developing embedded programs.
7.
 - a) What are the analogies between process, task and thread? Also list the difference between the process, task and thread.
 - b) Explain the term multi tasking OS and multi tasking scheduler.
8. Write short notes on any two of the following.
 - a) Task synchronization
 - b) Interrupts of 8085
 - c) Programming in Embedded C
